

Optimization

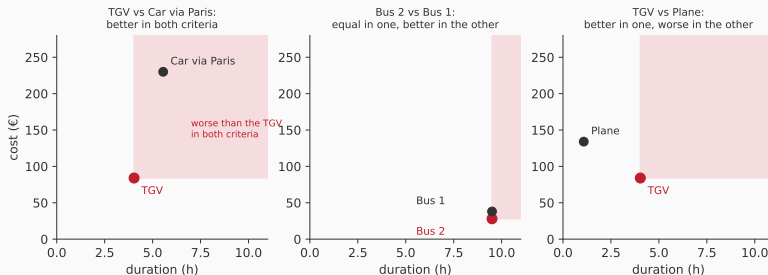
6.2 Pareto front

José Firmino Aguilar Madeira

From intuition to definition: what is the rule?

6.1 ended with the question: how do we define that one solution is unequivocally worse than another?

Three Nantes–Lille pairs:

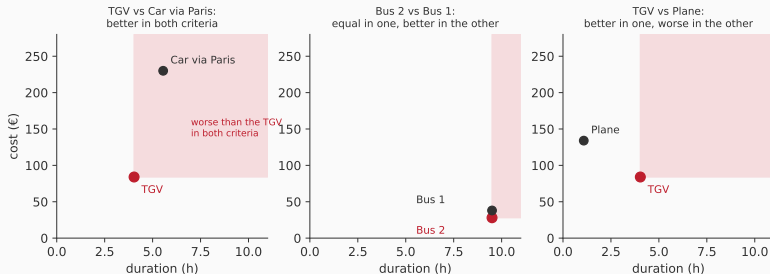


In which of these cases can we eliminate an option without knowing the decision-maker's preferences?

From intuition to definition: what is the rule?

6.1 ended with the question: how do we define that one solution is unequivocally worse than another?

Three Nantes–Lille pairs:



In the first two, yes: the TGV is better in everything; bus 2 takes *the same* time and costs less — a tie in one criterion and a win in the other is also enough.

In the third, no: the plane is faster and the TGV cheaper. **Neither is better in both:** without additional preferences, they cannot be ordered.

Pareto dominance

For $\mathbf{z}, \mathbf{z}' \in \mathbb{R}^m$ (minimization):

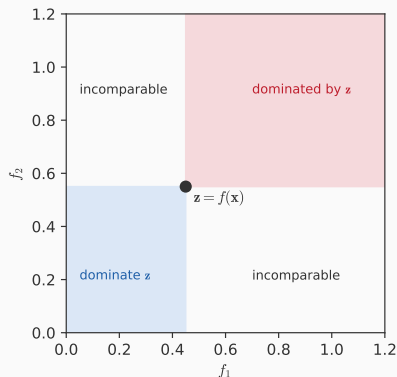
Definition

$\mathbf{z}$ **dominates** $\mathbf{z}'$, $\mathbf{z} \prec \mathbf{z}'$, if

$$z_i \leq z'_i \text{ for all } i \quad \text{and} \quad z_j < z'_j \text{ for some } j :$$

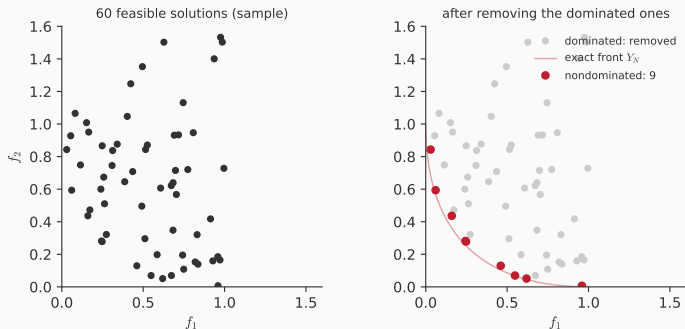
no worse in any objective and better in at least one.

- $\mathbf{x}$ dominates $\mathbf{x}'$ if $\mathbf{f}(\mathbf{x}) \prec \mathbf{f}(\mathbf{x}')$.
- TGV $\prec$ car via Paris; bus 2 $\prec$ bus 1; TGV and plane: **incomparable**.
- It is a **partial order**: there are pairs we cannot order. The decision-maker decides between them.



Relative to $\mathbf{z}$: **south-west**, those that dominate it; **north-east**, those it dominates; in the other two quadrants, the incomparable ones.

Removing the dominated: the front appears



60 feasible solutions of the example from 6.1; once the dominated ones are removed, 9 remain, all close to the red curve. With *all* the feasible solutions, what remains has a name: the **Pareto front**.

The front is what is left of Y after removing everything that is dominated.

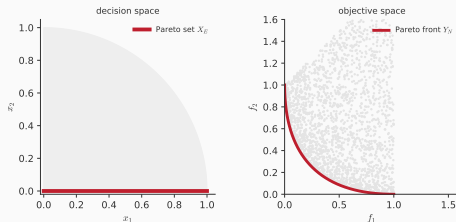
Pareto set and Pareto front

$\mathbf{x}^* \in X$ is **efficient** if no feasible solution dominates it:

$$\mathbf{x}^* \in X_E \iff \nexists \mathbf{x} \in X : \mathbf{f}(\mathbf{x}) \prec \mathbf{f}(\mathbf{x}^*).$$

- **In the decisions:** $X_E \subset X$, the **Pareto set** — the efficient designs.
- **In the objectives:** $Y_N = \mathbf{f}(X_E)$, the **Pareto front** — the performance of those designs.

In the example: $X_E = \{(x_1, 0) : 0 \leq x_1 \leq 1\}$; Y_N : $f_2 = (1 - \sqrt{f_1})^2$.



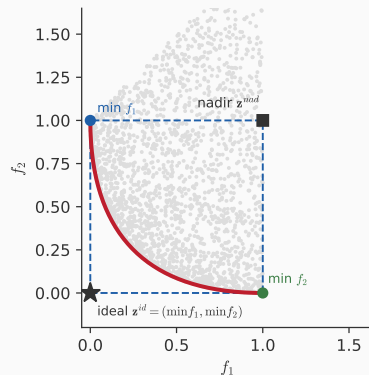
Pareto set $\neq$ Pareto front

For later (6.3, 6.4): $\mathbf{x}^*$ is *weakly efficient* if there is no solution strictly better in *all* objectives.

Ideal and nadir: the box that contains the front

Ideal. If we could choose the best of each objective *separately*: $z_i^{id} = \min_{\mathbf{x} \in X} f_i(\mathbf{x})$.

In general no solution attains all these values at the same time: z^{id} is generally unattainable. Here, $(0, 0)$ is not the image of any $\mathbf{x}$.



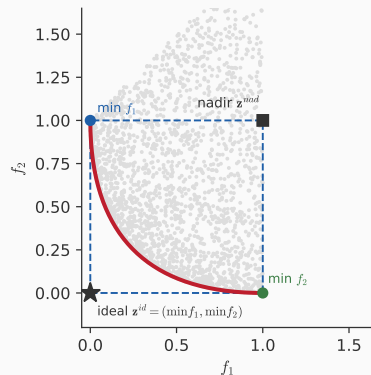
In practice an estimate $\hat{z}^{nad}$ is often used (for example, over the approximate front).

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Nadir. The worst value of each objective *among the efficient solutions*: $z_i^{nad} = \max_{\mathbf{x} \in X_E} f_i(\mathbf{x})$.



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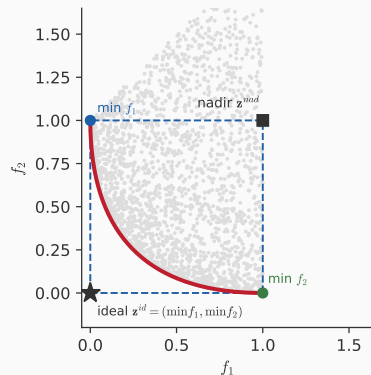
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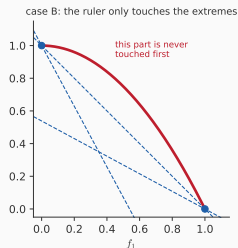
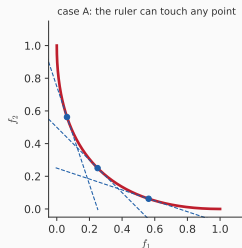
They serve to **delimit** the relevant region and to **normalize** objectives with different scales (6.3).



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The geometry of the problem: convex or not?

Why do we need to know? Because it will determine which points the weighted sum can find (6.3).



A ruler (straight line) pushed against the front, coming from below and from the left.

Case A: it can touch **any** point first.

Case B (the nonconvex example, below): there is a region that **no ruler can touch first**. (The front can also be discontinuous or discrete.)

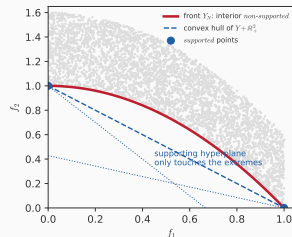
Nonconvex example of the chapter: $\min(x_1, 1 - x_1^2 + x_2)$, $\mathbf{x} \in [0, 1] \times [0, 0.6]$;
 $X_E = \{(x_1, 0) : 0 \leq x_1 \leq 1\}$; $Y_N: f_2 = 1 - f_1^2$.

Strictly speaking, “convex” refers to the upper image $Y + \mathbb{R}_+^m$, not to the curve Y_N on its own.

Supported and non-supported efficient points

A ruler with slope given by the weights (w_1, w_2) : it moves in from below and from the left until it touches the attainable region.

- Where it touches: a *supported* point.
- The part of the front that the ruler can never touch first: *non-supported*.



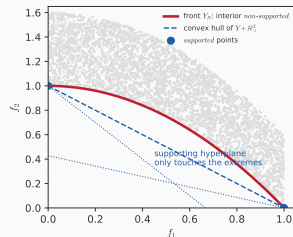
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Definition

$\mathbf{z} \in Y_N$ is *supported* if it minimizes $\mathbf{w}^T \mathbf{z}'$ over Y for some weights $w_i > 0$, $\sum_i w_i = 1$. Otherwise, *non-supported*.



On the nonconvex front (above) only the two extremes are *supported*. In a convex problem all the efficient points are, except possibly limit points that require a zero weight (the extremes of the convex example).

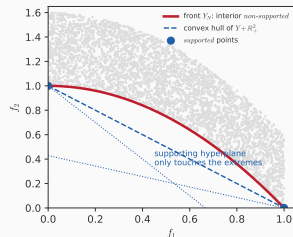
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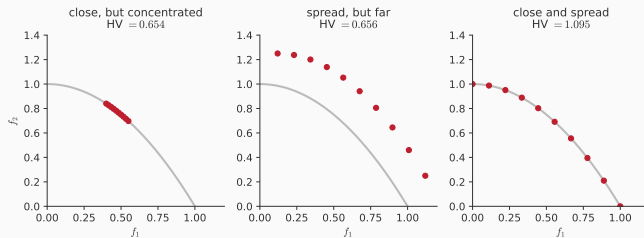


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Why it matters

The *weighted sum* (6.3) only finds *supported* points; the *ϵ -constraint* (6.4) also reaches the *non-supported* ones.

Two different goals: convergence and diversity



HV (hypervolume, next page, with $\mathbf{r} = (1.1, 1.65)$): 0.654, 0.656 and 1.095, from left to right.

An *a posteriori* method returns a finite set of points. We want:

convergence: get **close** to the front;

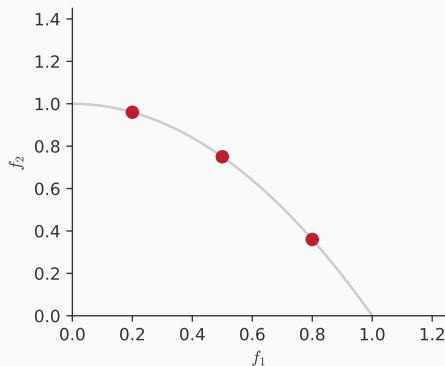
diversity: **cover it** well.

The first two approximations each fail one of the goals; the third meets both.

This is what DMS aims for with the list and the ordering (6.5), and NSGA-II with rank and crowding (6.6).

Measuring quality: the hypervolume

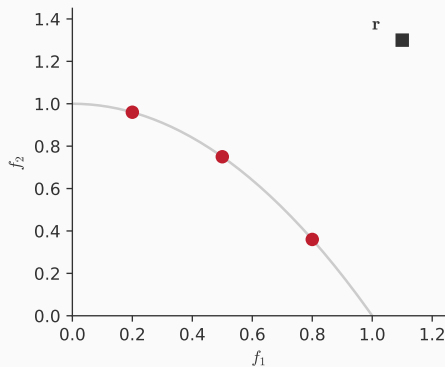
Three nondominated points.



$\mathbf{r} = (1.1, 1.3)$, chosen for illustration. Other indicators: GD, IGD, spread.

Measuring quality: the hypervolume

A **reference point** $\mathbf{r}$, worse than all of them.



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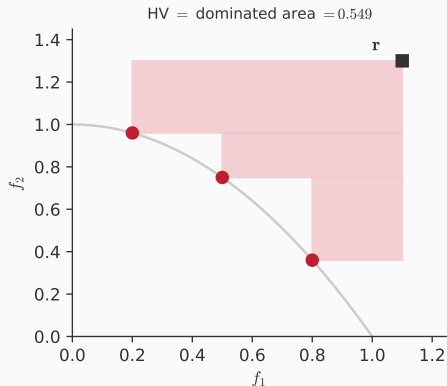
Measuring quality: the hypervolume

The region dominated by the points, bounded by $\mathbf{r}$:

$$HV = \text{dominated area (in 2D)}$$

- better **convergence** tends to increase the area;
- better front **coverage** tends to increase the area.

The HV **combines information on both goals**.



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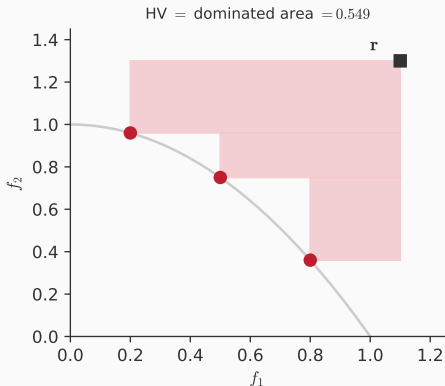
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To compare methods

Same problem, $\mathbf{r}$ and budget n_f (n_f = evaluations of the vector $\mathbf{f}$); stochastic methods: several runs, median and quartiles (3.2.1; Ch. 5). The value depends on $\mathbf{r}$ and on the objectives' scale.



$\mathbf{r} = (1.1, 1.3)$, chosen for illustration. Other indicators: GD, IGD, spread.

So, what does it mean to “solve” a multiobjective problem?

One objective

$$\mathbf{x}^*$$

a minimizer

Several objectives

$$Y_N$$

or, numerically,

$$A \approx Y_N$$

a finite set of points

A good approximation A :

convergence + diversity

KEY TAKEAWAYS — 6.2

1. **Dominance:** $\mathbf{z} \prec \mathbf{z}'$ if it is not worse in any objective and is better in at least one. Some pairs are incomparable.
2. **Pareto front:** what remains after removing all the dominated points. Pareto set X_E (decisions) $\neq$ front Y_N (objectives).
3. **Ideal and nadir:** the box that contains the front; they are used for normalization.
4. **Supported vs. non-supported:** on a nonconvex front there are points that no weighted sum reaches.
5. **Convergence and diversity** are different goals; the **hypervolume** combines both into a single indicator.

We now know which object to compute: the front. **How do we compute it?**

First idea: turn several objectives into a single one again $\rightarrow$ **6.3 Aggregation**

References Full list: *Course bibliography*, at the end of the course PDF.

- Miettinen, K., *Nonlinear Multiobjective Optimization*, Kluwer Academic Publishers, 1999.
- Jahn, J., *Vector Optimization: Theory, Applications, and Extensions*, 2nd ed., Springer, 2011 — Ch. 4.

Resources: —